

Computing Educational Activities involving PEOPLE rather than THINGS appeal more to Girls (School Perspective)

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ABSTRACT

We investigate gender-differential preferences for computing activities in primary and lower secondary education. We conducted an experimental study with $N=815$ pupils from 2nd, 5th, and 8th grade across five schools. Pupils chose between isomorphic PEOPLE vs. THINGS versions of four hypothetical computing-related tasks. Results show no gender difference in 2nd grade, but significant differences in 5th grade and 8th grade, with girls consistently expressing a stronger preference for PEOPLE-oriented activities than boys. At the same time, in absolute terms, both boys and girls exhibited an overall preference for the PEOPLE versions of the tasks. Extending earlier findings to a school context, we suggest that, already by around 5th grade, computing educators may consider favoring PEOPLE- over THINGS-oriented themes, as these appear to offer a more inclusive appeal to pupils in school.

CCS CONCEPTS

• **Social and professional topics** → **Computing education.**

KEYWORDS

Gender, People–Things, Thematic framing, Primary education, Educational experiments

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1 INTRODUCTION

The gender imbalance in computing remains one of the most persistent challenges in the field. Despite decades of targeted initiatives, women are still significantly underrepresented among information and communication technology (ICT) specialists. In the European Union, for example, only *one in five* ICT specialists in 2025 were women (19.5%) [9], while just *one in six* employees with an ICT education were women (16.6%) [10]. This imbalance has consequences not only for the diversity and inclusivity of the discipline, but also for society's ability to meet the continuing demand for ICT professionals. Despite rapid advances in AI, shortages of ICT specialists remain widespread across Europe [7].

Beyond the labour market, computing education has value as a component of modern literacy. Computational thinking has been associated with broader cognitive benefits including problem solving and analytical reasoning [14, 15, 36]. Consequently, increasing participation in computing education benefits not only the future ICT workforce, but society more generally.

One possible explanation for persistent gender differences is that seemingly superficial aspects of educational activities influence how appealing they appear to learners: Research on the PEOPLE–THINGS dimension [16, 17] suggests that women and men (girls and boys) differ, on average, in their *preference* for activities involving PEOPLE versus THINGS. Recent computing education research has shown that merely changing the *thematic framing* of otherwise isomorphic computing tasks can influence students' preferences in upper secondary school [5] and university settings [19]. This raises a natural question: *when do these gender differences first emerge?*

Answering this question requires looking to primary school, where many pupils first encounter computing-related concepts. If gender-differential preferences are already detectable at this stage, relatively small design choices in educational materials may influence how inclusive computing education feels long before pupils decide whether computing is “for them.”

As of 2026, Denmark still lacks *mandatory* computing education in primary and lower secondary school. Following a 2018–21 pilot, the government decided that from 2027/28 some computing elements will be integrated into selected subjects from 1st grade, while computing itself will be offered only as an *elective* in 7th–8th

grade. This decision has been widely criticised for reinforcing gender and socio-economic divides. A teacher with experience from a 9th-grade elective notes [31]:

“In my experience, it is mostly boys who are already skilled at [digital] technology that choose the subject.”

— [Lis Zacho, University College Copenhagen]

The chair of the national expert group similarly argues [31]:

“In order to ensure that all children, regardless of socio-economic background and gender, learn the necessary tools to navigate a digital future, the subject should be mandatory instead of elective; otherwise, especially girls risk being left behind.”

— [Ole Sejr Iversen, Professor, Aarhus University]

Motivated by these concerns, we investigate whether gender-differential preferences for PEOPLE- versus THINGS-oriented computing activities are already observable during primary and lower secondary education. We make the following contributions:

- (1) **Extension to school-aged pupils.** We extend prior PEOPLE–THINGS research from upper secondary and university education to primary and lower secondary school.
- (2) **Evidence of early emergence.** We provide empirical evidence that gender-differential preferences become statistically detectable by 5th grade within our experimental setting.
- (3) **Developmentally calibrated methodology.** We introduce a developmentally calibrated experimental design for computing education spanning ISCED levels 1 and 2 that enables systematic investigation of PEOPLE–THINGS preferences across different stages of schooling.

2 BACKGROUND AND RELATED WORK

The underrepresentation of women in computing is well studied [11, 24, 29]. The PEOPLE–THINGS dimension has long been proposed as one explanation for gender differences in vocational interests. In a series of studies, Lippa [16] demonstrated that women, on average, show greater interest in occupations involving PEOPLE, whereas men tend to show greater interest in occupations involving THINGS. These findings have subsequently been corroborated and extended in numerous studies and meta-analyses [17, 33, 34]. In particular, Su and Rounds [33] found that gender disparities across STEM disciplines are better explained by the degree to which fields involve working with PEOPLE versus THINGS than by the level of quantitative ability they require. Furthermore, orientations along the PEOPLE–THINGS dimension have been shown to predict educational preferences, choice of major, and retention in THINGS-oriented fields such as science and engineering [37].

Although the framework originated in the context of career choice, it also provides a useful perspective on educational design. If learners respond differently to activities depending on whether they are framed around PEOPLE or THINGS, then educational activities may become more appealing through changes to thematic framing alone, while leaving the underlying technical content unchanged.

Recent computing education research has begun to investigate whether the PEOPLE–THINGS dimension also manifests itself in relation to educational material. A recent study [5] extended Lippa’s hypothesis to upper secondary computing education using pairs

of isomorphic hypothetical computing tasks differing only in their thematic framing. It found that women had 2.7 times higher odds than men of preferring a PEOPLE-themed task over an isomorphic THINGS-themed task. Furthermore, the effect was amplified by prior programming experience: a woman *without* programming experience had 3.8 times higher odds of preferring the PEOPLE version than a man *with* programming experience. Similar gender-differential preferences were subsequently observed among first-semester university students in computing through mandatory programming assignments [19] and across entire educational programmes [12]. Together, these studies demonstrate that the PEOPLE–THINGS dimension also influences learners’ preferences within computing education.

Prior work further suggests that seemingly superficial characteristics of learning environments can influence students’ engagement with computing. Cheryan et al. [4] demonstrated that stereotypical environmental cues affect students’ interest even when the underlying activity remains unchanged. Margolis and Fisher [20] argue that many women are drawn to computing through its potential to solve real-world problems and help people. Postner et al. [30] observe women leaning towards humanitarian applications.

Gender stereotypes concerning computing interests also emerge early and contribute to differences in interest and sense of belonging [22]. Collectively, these findings indicate that relatively small design choices—including examples, contexts, wording, and thematic framing—may shape students’ motivation, engagement, and sense of belonging.

Even though it has been established that gender differences in interests surface quite early [13, 22] and that the PEOPLE–THINGS framing effects are visible in upper secondary and university education, much less is known about framing effects in compulsory lower education have primarily investigated general interest in programming or computational thinking rather than thematic framing [2, 13, 18, 23].

Our study addresses this gap by extending existing findings on thematic framing to explore when the gender-differential preferences toward PEOPLE vs. THINGS becomes detectable.

3 METHODOLOGY

3.1 Objectives

We consider computing education tasks in primary and lower secondary school (ISCED levels 1 and 2). We investigate the gender-differential preferences along the PEOPLE–THINGS spectrum at three different points during primary and lower secondary school:

RQ1 (Gender Differences):

Do girls differ from boys in their preferences for PEOPLE vs THINGS-oriented computing activities in any of 2nd, 5th, and 8th grade?

RQ2 (Absolute Preferences):

How appealing are PEOPLE- vs THINGS-oriented computing activities to girls and boys in absolute terms?

For 2nd graders—who lack prior exposure to formal computing—the concept was broadened from *computing* to *science* at large (STEM).

3.2 Educational Context

The vast majority (around 80%) of children in Denmark attend a tuition-free government-funded public school [27] for the mandatory ten years of primary and lower secondary education (ISCED levels 1 and 2). Our study was conducted in five such public schools. All schools follow the national curriculum, which specifies competence, skills, and knowledge goals for each subject [28], while leaving teachers some freedom in implementation. The average class size is 20 pupils [8]. The school system is traditionally divided into three main stages: grades 0–3, grades 4–7, and grades 8–9 (and an optional 10th grade). At the end of 9th grade, pupils take a school-leaving exam. Our study has selected a grade from each stage. The typical age of pupils is 8–9 years in 2nd grade, 11–12 years in 5th grade, and 14–15 years in 8th grade. We note that the onset of puberty falls around grades 3–5 for girls and around grades 5–7 for boys [35].

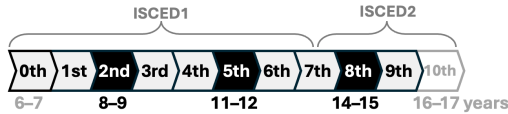


Figure 1: Overview of the school system in Denmark with the participating grades (2nd, 5th, and 8th) highlighted.

3.3 Subject Selection

Grades. We selected a grade from each main stage of the education system. We avoided 1st grade children because we deem them too young to meaningfully participate in our study; many of them will not be able to read yet and may not know how to fill in the questionnaire. We also avoided 9th grade since they would be busy preparing for the school-leaving exam.

Schools. With the help from educational research colleagues, we managed to enlist five schools geographically distributed around the country (roughly 100 kilometres apart) from the four most populous cities in the country. According to the World Bank, Denmark is predominantly urban with 88% living in cities [3].

3.4 Hypothetical Tasks

We selected four qualitatively different educational tasks representative of what pupils might be exposed to in a computing class during primary and lower secondary school:

- A: *writing* an article on digital technology
- B: *collecting* and *visualizing* data
- C: *designing* an App
- D: *programming* an App

For younger grades, the tasks are simplified to be age-appropriate; e.g., the 8th grade tasks of *writing* (an article), *collecting/visualizing* (data), *designing* (an App) and *programming* (an App) becomes *reading* (an article), *counting* (data), *drawing* (an App), and *counting* (something) for 2nd grade. Since the pupils cannot be expected to know about computing and programming, all educational activities were made to be *hypothetical* in the sense that they did not actually have to solve the tasks, but merely choose which of *two versions*

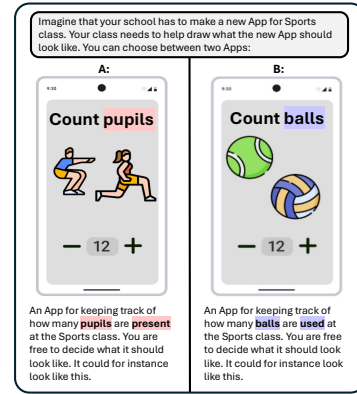


Figure 2: Exercise 2c (2nd grade, Exercise c). Shared introduction at the top, PEOPLE left, and THINGS right. Thematic treatment differences highlighted as: PEOPLE vs THINGS.

keeping	track	of	how	many	pupils	are	present
keeping	track	of	how	many	balls	are	used
verb	noun	prep	adverb	quantifier	noun	verb	adjective
gerund	sing.		interrog.	determiner	plural	aux.	predicate

Figure 3: Excerpt from Exercise 2c illustrating syntactic isomorphism between the PEOPLE (top) and THINGS (bottom) versions. Thematic differences are highlighted in red / blue.

(elaborated below) of the task they would prefer to do. The two versions were introduced by a common description, e.g.

Imagine your class has had a school subject where you learned how to program Apps. You need to make a project which is about programming an App.

3.5 Treatment

As treatment, we created *two isomorphic versions* of all (four) tasks: a PEOPLE-themed version and a THINGS-themed version. This gives rise to four exercises, {A, B, C, D}, pitching the two versions against each other. At the top of an exercise, there is a shared description introducing the hypothetical task. Below that, the two versions are shown side by side. Both versions will have an image above a textual description. As an example, Figure 2 shows Exercise c for 2nd grade (2c).

For all 12 exercises, $\{2, 5, 8\} \times \{A, B, C, D\}$, the images and textual descriptions are *syntactically*, *semantically*, and *compositionally isomorphic* and vary only in terms of the thematic treatment (in that one is about PEOPLE, the other about THINGS):

Syntactically, we made sure that the versions had the same grammatical structure. Figure 3 features an excerpt from Exercise 2c and illustrates how sentences have similar syntactic sentence structure, with differences limited to individual words (highlighted in color); e.g., *pupils* vs *balls* (both nouns, in plural form) or *present* vs *used* (both adjective predicates).

Semantically, the versions are also similar in that the overall scenario of the exercise is the same for both versions. In Exercise 2c (Figure 2), both versions are about “keeping track of something”

Table 1: Characterization of the four exercises {A, B, C, D}.

	Subject	Activity	Scenario	Treatment (theme):	
				PEOPLE	THINGS
A	Sci/Tech	Article	Traveling	Astronaut	Spaceship
B	Math	Calculations	Shopping	Customers	Packets
C	Sports	Application	Organization	Pupils	Balls
D	Math	Project	Calculation	Heartbeat	Windmill

Table 2: Topics of the {2nd, 5th, 8th} × {A, B, C, D} exercises.

	2nd	5th	8th
A	Science	Computing	Computing
B	Mathematics	Computing [†]	Data Science
C	Computing [†]	Computing [†]	Computing [†]
D	Mathematics	Data Science	Computing [†]

[†]) Computing exercise involving App development.

(either *pupils* or *balls*) with a certain property (either *present* or *used*).

Compositionally, the images are also similar both in terms of *colors* (black font on gray background with yellow/orange iconic figures) and *composition* (headline text above two images side-by-side above a display, both showing the number 12, surrounded by decrease and increase buttons $-/+$, all inside a mobile cell phone).

Table 1 provides a characterization of the four exercises (A–D) in terms of task/activity, overall scenario, and the themes of the two (PEOPLE vs THINGS) variants. The resulting 12 exercises used in our study are shown in Figure 7 (see supplementary material for larger versions).

All exercises are aimed at the particular age range using the Common Objectives of Technology Comprehension. Exercises for 2nd grade (8–9 year olds) will thus be simple with limited textual descriptions (as in Figure 2). Table 2 lists the topics of the exercises; most exercises gravitate around *computing* (including *data science*) and many involve App development. Due to limited concept differentiation and familiarity with the digital realm in younger children, the concept (discipline) of *computing* (or *data science*) may generalize to the wider concept (discipline) of *science* or *mathematics* for some of the 2nd grade exercises. Exercises for older children in 8th grade (14–15 year olds) will be significantly more elaborated and feature longer textual descriptions.

Table 3 shows evidence of this age progression from 2nd to 8th grade: We see that the exercises use progressively *more* words (from 61 to 97 to 124) and *longer* words (from 4.4 to 4.6 to 4.8 average word length) in 2nd, 5th, respectively, 8th grade. See Figure 7 or the supplementary material for the full set of exercises used in our study.

Rating scale. All exercises are accompanied by a question on a preference scale from 1–5; the scale quantifies the preference for a particular version over the alternative and includes the possibility of indifference (see Table 4).

Task design and validation. The exercises were iteratively constructed by members of the research team, drawing on design principles and thematic structures established in prior work on the PEOPLE–THINGS dimension [5, 19]. Particular care was taken to

Table 3: Average number of words and average word length per exercise for each of the three grades, including a breakdown into the three main elements of an exercise.

	Avg. word count			Avg. word length		
	2nd	5th	8th	2nd	5th	8th
Shared introduction	25	31	36	4.2	4.4	4.3
PEOPLE-version	18	33	44	4.6	4.5	4.8
THINGS-version	18	33	44	4.6	4.8	5.1
Entire exercise	61	97	124	4.4	4.6	4.8

Table 4: Answer options: preference score 1–5.

Score	Explanation
1	I would much rather do $\langle Left \rangle$
2	I would rather do $\langle Left \rangle$
3	I have no preference for neither $\langle Left \rangle$ nor $\langle Right \rangle$
4	I would rather do $\langle Right \rangle$
5	I would much rather do $\langle Right \rangle$

ensure syntactic and semantic isomorphism between the PEOPLE- vs. THINGS-versions of each task, so that only thematic framing differed. The materials were reviewed multiple times for conceptual clarity and educational appropriateness, including input from researchers with expertise in education and philosophy. Prior to the main study, the exercises were piloted with two pupils from 4th and 6th grade, leading to minor revisions of wording and presentation to ensure comprehension.

3.6 Experiment Design

We now present the details of the experimental design aside from the exercises intended for the participating pupils.

Facilitation. Three of the authors served as experimenters and undertook one-day experiments at each of the five participating schools during the Spring of 2025. At each school, a so-called *coordinator* facilitated experimental execution by contacting teachers of relevant classes and setting up one-day schedules which sometimes involved parallel execution in multiple classes using multiple experimenters. For simplicity, the tests were conducted in class, on paper (eliminating potential problems with technology) during conventional 45 minute teaching lessons where the local teacher would be responsible for keeping order.

Reading. Since a recent study [25] demonstrated that about a *quarter* of 4th grade pupils in Denmark have reading difficulties, we decided to use voice recorded audio playback for the 2nd and 5th grades. Using pre-recordings ensures similar reading of both variants (PEOPLE & THINGS) without any unintended attention given to either variant; e.g., intonation, pausation, or inflection. By 8th grade, the number of challenged readers are lower [26]; hence, they would read the questionnaire themselves at their own pace.

Variants. To account for left-to-right reading bias and the risk that pupils might arbitrarily pick the first (i.e. leftmost) version, we created two *variants* of the questionnaire with alternating sequences of which version was to the left vs right.

Demonstration. To make sure pupils, especially younger ones, knew what to do and how to answer the questions, we made three

demonstration exercises involving choosing between: (a) *lasagna* vs *pizza*; (b) *lion* vs *bear*; and (c) *pirate* vs *vampire*. For each of the demo exercises, the pupils were shown qualitatively different kinds of answers: (a) *strong* preference for the *left* variant (a score of 1 for *lasagna*); (b) *weak* preference for the *right* variant (a score of 4 for *bear*); and (c) *indifference* (a score of 3 meaning equal preference for *pirate* and *vampire*); cf. the rating scale (Table 4).

3.7 Ethical Considerations

The study received formal approval from the Ethical Board of our university on behalf of the Head of Research based on a description of the intended study; in particular, its methodology. The method is the same as an earlier study [5] (published in ACM ICER 2021), except for taking place in the context of *primary and lower secondary* rather than *upper secondary* education. The study was approved by the coordinator of each participating school as well as the individual teachers of each participating class. Teachers would inform parents about the experiment via AULA (a national platform undertaking all communication between parents and school). Pupils were explicitly informed that participation was not mandatory in that they did not need to complete the (entire) questionnaire. Gender was solicited in an open-text field where the pupils could write *girl*, *boy*, or *anything else*. We deliberately did not collect names or anything else that could identify individual pupils. The study was conducted on paper and was 100% anonymous and cannot, in any way, be traced back to any particular student. Since it is not just pseudo-anonymous, the Ethical Board of our University ruled that it is in compliance with relevant regulations (including GDPR¹).

3.8 Statistical Methods

For consistent interpretation, all responses are relabeled so that 1 indicates a strong preference for the PEOPLE version and 5 a strong preference for THINGS.

We use a cumulative logit model with a random intercept [1], which is a standard statistical model for multinomial data with an ordinal response variable (the score 1–5 given to an exercise) and repeated measurements (multiple exercises are rated by the same pupil). The same model was used for the statistical analysis of a similar experimental setup for upper secondary education [5]. All analyses are carried out in R using the `cglm2` framework for fitting cumulative link mixed models as provided by the `ordinal` package [6].

Although based on the same context, the four exercises A–D are not directly comparable across the three grades, and so each grade has been analyzed separately.

We are interested in quantifying the probabilities with which a pupil assigns each of the scores 1–5 to a given exercise. For instance, the set of probabilities (0.1, 0.1, 0.4, 0.3, 0.1) would mean that the pupil has a probability of 0.4 of rating the exercise with a 3 (indifferent), and is more prone to giving a score of 4 (probability 0.3) than a score of 2 (probability 0.1). For the purpose of our study, we are particularly interested in establishing whether this distribution of scores is significantly different for a boy than for a girl.

The probabilities of the five scores may instead be summarized through four cumulative probabilities: Imagine that we split the

rating scale between two categories j and $j+1$. We can then consider the probability of choosing a score between 1 and j (*in the direction of PEOPLE*) rather than a score between $j+1$ and 5 (*in the direction of THINGS*). Considering the above example distribution of (0.1, 0.1, 0.4, 0.3, 0.1) and a split between categories 4 and 5, the pupil has a probability $0.1 + 0.1 + 0.4 + 0.3 = 0.9$ of rating somewhere in 1–4 (the direction of PEOPLE). From this example it is clear that, depending on *where* the rating scale is split in two, the terminology of rating *in the direction of PEOPLE* may actually cover also a slight preference for THINGS as this is more in the direction of PEOPLE than is a strong preference for THINGS. The set of four cumulative probabilities would be (0.1, 0.2, 0.6, 0.9), corresponding to groups of scores 1, 1–2, 1–3, and 1–4.

The cumulative logit model is formulated in terms of *odds* of rating in the direction of people, $P(\text{score} \leq j)/P(\text{score} > j)$, rather than the probabilities $P(\text{score} \leq j)$. In the context of our study, these odds describe how many times more likely the person is to rate somewhere in the direction of PEOPLE than somewhere in the direction of THINGS. The above pupil has odds $0.9/(1 - 0.9) = 9$ of rating somewhere in 1–4 (the direction of PEOPLE) rather than 5 and odds $0.2/(1 - 0.2) = 4$ of rating in 1–2 rather than 3–5.

Quantifying effects via odds ratios. Any difference between boys and girls in terms of the distribution of scores may then be quantified in terms of *odds ratios*, which express how many times higher the odds of rating in the direction of PEOPLE are for a girl than for a boy.

Model specification. For each score $j \in \{1, 2, 3, 4\}$, the model describes for a pupil the (log-)odds of rating in the direction of PEOPLE, i.e. the (log-)odds of giving a score of j or lower. For all grades, we have used as a baseline model the least restrictive model structure, which lets odds vary freely over all combinations of gender and exercise as well as over the four possible splits of the rating scale. We included also a pupil-specific effect (a *random intercept*) to account for the dependence between ratings of a specific pupil. This effect may also be interpreted as modeling a variation between pupils in how strongly they tend to state their preference. The pupil-specific effect is assumed to be Gaussian with mean 0 and an unknown variance, which is then estimated from the data in our study. The least restrictive model allows for a potentially different gender effect (odds ratio) for every combination of exercise and split between categories. For the sake of parsimony and interpretability, we have carried out a statistical model reduction with the principle of reducing to the simplest possible model structure that data supports and includes gender as a factor.

4 RESULTS

The experiment was conducted during the spring of 2025 and resulted in the participation of $N=822$ pupils from the five participating schools. Table 5 shows a breakdown of pupils according to grades and school. 2nd grade was represented by 284 participating pupils, 5th grade by 274, and 8th by 264. To reduce potential familiarity bias, one 2nd grade pupil related to an author was excluded from the analysis. Two pupils were omitted from the study for responding (appropriately) with “non binary” as their gender. (The PEOPLE–THINGS gender correlation does, unfortunately, not incorporate the notion of *non-binary* [16].) Four pupils did

¹General Data Protection Regulation: European law protecting personal data in EU.

Table 5: Number of pupils participating in the experiment according to school & grade. †) Langeskov is 10km from Odense. The total (N=822) reflects all participating pupils; analyses are based on N=815 after exclusions, as explained in Section 4.

City	School	Size	2nd	5th	8th	Total
Aalborg	Sofiendalsskolen	640	39	68	52	159
Odense	Langeskov skole†	597	54	46	37	137
Aarhus	Gammelgaardsskolen	987	98	92	100	290
Copenhagen	Dyvekeskolen	517	-	68	-	68
	Løjtegårdsskolen	759	93	-	75	168
Total		3,500	284	274	264	822

not meaningfully answer the gender question and were therefore excluded from the study (e.g., “Dishwasher” and “Lego-man”). Ultimately, the study is based on **N=815** pupils with a gender ratio close to 50/50, as expected, with 50.7% girls & 49.3% boys (comprising 414 girls & 401 boys). All but two of the 815 included pupils rated all four exercises; one 2nd grade girl did not rate 2A, and one 8th grade boy only rated 8c. The average age, reported as an integer, by the pupils in 2nd, 5th, and 8th grade was 8.4, 11.4, respectively, 14.4 which is to be expected for a spring-time experiment.

Task exclusion (8c). Unintentionally, the THINGS-variant of Exercise 8c (Exercise c for 8th grade) erroneously featured the word “activity”. Ontologically, an *activity* usually implies *agents* (with *agency*) and is typically associated with *social interaction*, making the formulation closer to the PEOPLE side of the spectrum. For this reason, we exclude all data relating to this exercise.

Preference data (2nd, 5th, and 8th grade). Figure 4 shows the empirical distribution of the ratings in each age group, aggregated over all exercises. This simple overview of the collected data is consistent with our anticipation at the outset of this study: **For 2nd grade**, we see no difference between boys and girls, and there does not appear to be a preference for either exercise; indeed “indifferent” is the most common category chosen. **For 5th grade**, we see that more pupils indicate a preference for one or the other version, and that this preference among girls does appear to have a bearing towards PEOPLE, whereas there is no clear preference among the boys. **Among 8th graders** the preference for PEOPLE among girls seems slightly more pronounced than among the 5th graders. For comparison, Figure 4 includes the aggregated results from an earlier similar study in upper-secondary education [5]. There, we can see that both boys and girls expressed a tendency to prefer PEOPLE, but the girls more so than the boys. In Figure 5, where results are shown for each exercise separately, we get a much more nuanced picture of preferences, though the overall trends summarized by Figure 4 are maintained. From Figure 5 it is clear that preferences vary substantially between exercises and that this needs to be taken into account in the statistical analysis below.

4.1 Gender-Differential Preferences (RQ1)

We now turn to the results of a statistical analysis of the pupil preferences and the differences between boys and girls. As explained in Section 3.8, our model describes the odds of *rating in the direction of PEOPLE*, defined as giving a score j or less for a specific choice of score threshold $j \in \{1, 2, 3, 4\}$.

Gender Differences in 2nd Grade. For second graders, the effect of gender may be summarized in terms of a *single odds ratio across all four exercises* and it also does not depend on which score threshold is considered in defining the odds. The odds of a girl rating in the direction of PEOPLE are estimated to be 1.1 times higher than those of a boy, although the odds are not significantly different ($p = 0.31$). The support for equal odds (odds ratio of 1) is also reflected in the 95% confidence interval 0.9–1.4. The variation between pupils implies that the ratio of the odds of rating in the direction of PEOPLE for two specific pupils will vary depending on the extremity of their preferences. For 2nd graders, the *median* of odds ratios obtained by comparing a randomly picked girl to a randomly picked boy is 1.1, and 90% of odds ratios will be in the range 0.64–2.0. In summary:

OBSERVATION 1A (No gender difference in 2nd):

In **2nd grade**, there appears to be **no gender-differential interest** along the PEOPLE–THINGS spectrum: Girls and boys have the same odds of preferring the PEOPLE-version of an exercise.

Gender Differences in 5th Grade. For fifth graders, we see a significant gender effect ($p = 5 \cdot 10^{-6}$), although the pattern of preferences is slightly more complicated than for second graders. For exercises 5A, 5B, and 5D, we may summarize the gender effect for exercises by a single (median) odds ratio of 1.4: the odds of a girl rating in the direction of PEOPLE are 1.4 times higher than those for a boy. Exercise 5C exhibits a more complex pattern for the preferences. Here the gender effect depends on exactly which score threshold is considered in defining the odds, as Table 6 shows. Considering the odds of explicitly preferring PEOPLE (rating in $\{1, 2\}$), the median odds ratio of 1.2 is similar to that of exercises 5A, 5B, and 5D. In contrast, girls have around three times higher odds than boys of rating somewhere in $\{1, 2, 3, 4\}$ or $\{1, 2, 3\}$, but only around half the boys’ odds of giving a score of 1; indeed boys are more likely to give a score of 1 than the girls are. In summary:

OBSERVATION 1B (Gender difference in 5th grade):

In **5th grade**, we see a statistically significant gender difference in the preference in themes along the PEOPLE–THINGS spectrum.

Gender Differences in 8th Grade. For eighth graders, the effect of gender is significant ($p = 0.0016$) and can again be summarized in a single odds ratio. The (median) odds ratio is 1.6 with a confidence interval of (1.2, 2.1), and 90% of odds ratios comparing a randomly picked girl to a randomly picked boy would be in the range 0.64–4.0. In summary:

OBSERVATION 1C (Gender difference in 8th grade):

In **8th grade**, we see a gender difference in the preference along the PEOPLE–THINGS spectrum: the odds of a girl preferring an exercise involving PEOPLE are 1.6 times higher than those of a boy.

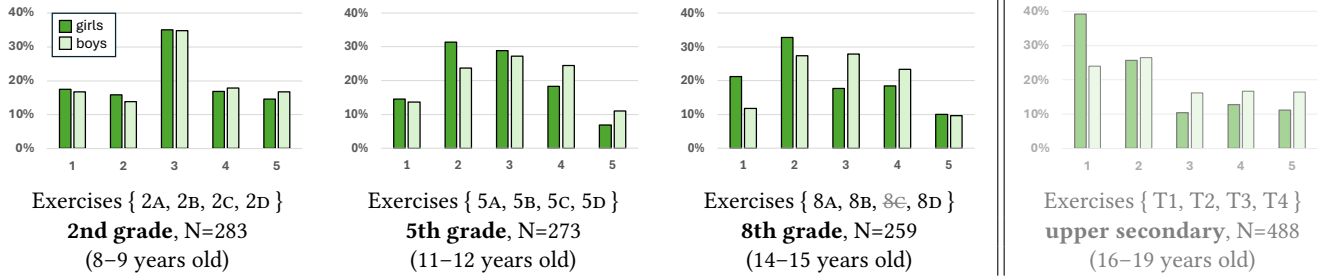


Figure 4: Aggregated results across exercises showing the distribution of pupil ratings. The rating scale goes from 1 (*strong PEOPLE* preference) over 3 (*indifference*) to 5 (*strong THINGS* preference). For comparison, the rightmost panel shows the corresponding results from an earlier and similar experiment for upper secondary education [5].

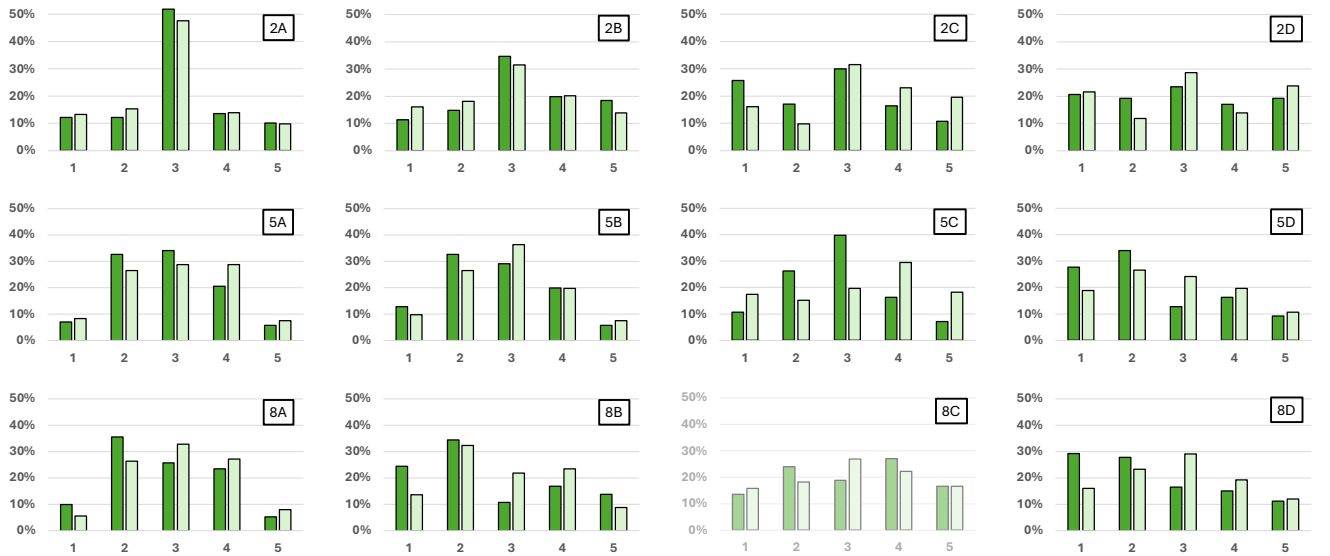


Figure 5: Pupil ratings by grade (rows) and exercise (columns) on a scale from 1 (*strong PEOPLE* preference) through 3 (*indifference*) to 5 (*strong THINGS* preference). Note: Although ratings for Exercise 8c are shown here, we remind the reader that they were omitted from the statistical analysis due to an ontological mistake in the wording of the exercise.

Table 6: Estimated gender effects quantifying how many times higher the odds of rating *in the direction of PEOPLE* are among girls than they are among boys. Median odds ratios are shown with 95% confidence intervals.

Grade	Exercises	Odds ratio	95% CI
2nd	2A–D	1.1	0.90 – 1.4
5th	5A, 5B, and 5D	1.4	1.1 – 1.9
	5c (odds of rating {1})	0.56	0.27 – 1.1
	5c (odds of rating {1, 2})	1.2	0.72 – 2.0
	5c (odds of rating {1, 2, 3})	3.1	1.8 – 5.3
	5c (odds of rating {1, 2, 3, 4})	2.9	1.3 – 6.5
8th	8A–D	1.6	1.2 – 2.1

4.2 Absolute Preferences (RQ2)

The gender *differences* in preferences, discussed above, do not say anything about the absolute preferences among boys or girls. Indeed it is possible that even though there is a higher preference for PEOPLE among girls than among boys, both genders may actually prefer the THINGS version. Looking at Figure 6, we get a detailed picture of the estimated probabilities of pupils preferring PEOPLE (rating 1 or 2), THINGS (rating 4 or 5), or being indifferent (rating 3). The preference distribution is shown for a median pupil (black, solid) as well as for a 5th percentile (grey, dotted) and a 95th percentile (grey, dashed) pupil.

We see that among second-grade pupils there is no clear preference for either version, and for exercise 2A pupils are actually most likely to be indifferent. Among fifth grade pupils there is more of a tendency to state an explicit preference, and for the girls it is

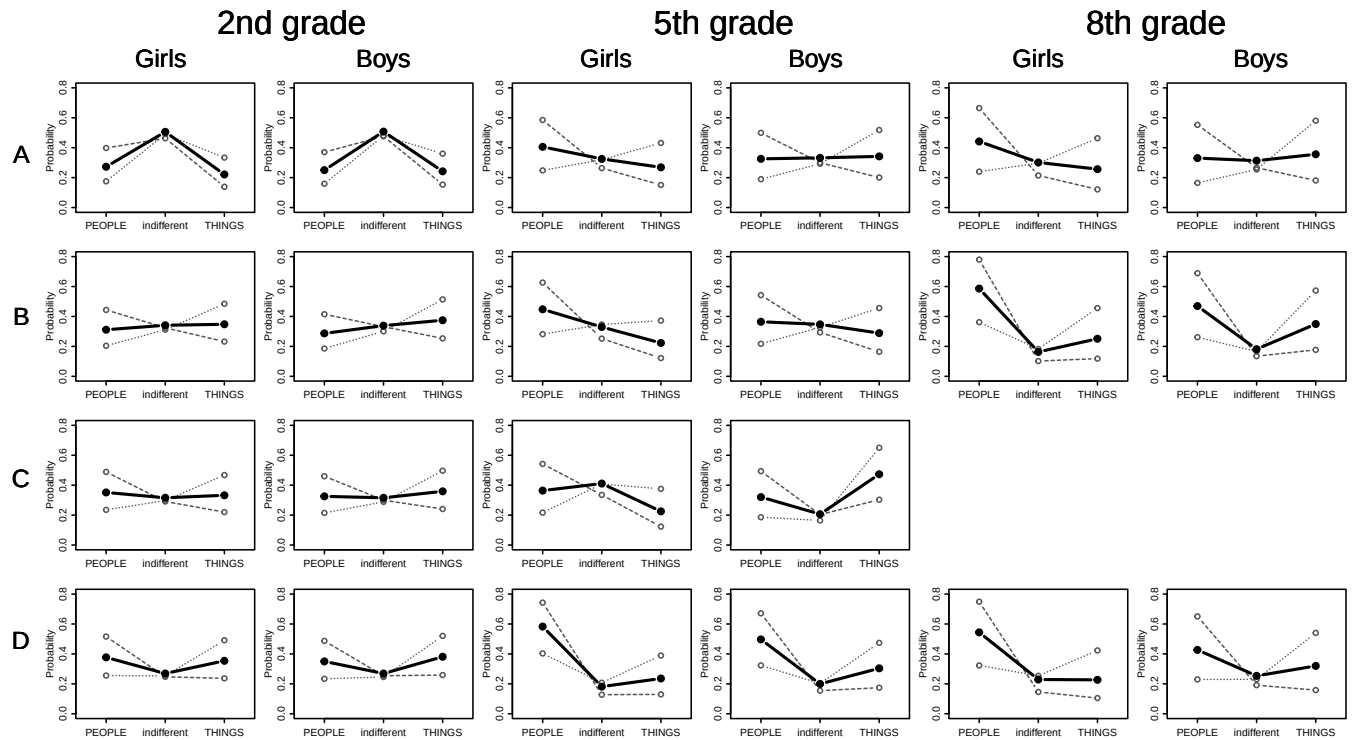


Figure 6: Estimated probabilities of selecting PEOPLE, indifferent, and THINGS for a median pupil (solid line), as well as for the 5th percentile (dotted) and 95th percentile (dashed) pupils.

clear that the probability of preferring PEOPLE is higher than preferring THINGS. The preference for PEOPLE is even more pronounced among girls in eighth grade. For several of the exercises we see that also boys are most likely to exhibit a preference for the PEOPLE version. In line with previous studies [5], we found a significant difference between exercises for all three grades in terms of the strength of pupil preferences.

OBSERVATION 2 (Absolute preferences):

For several of the exercises where preferences are clearly expressed, both girls and boys exhibit higher probability of preferring the PEOPLE– over the THINGS–version.

5 DISCUSSION

This study establishes that gender-differential preferences along the PEOPLE–THINGS dimension are detectable already in primary and lower secondary education, with indications that the effect becomes more pronounced over time.

Emergence of gender-differential preferences. Unlike previous studies in upper secondary and university computing education, our study asks when gender-differential preferences first become empirically detectable during schooling. Within our experimental framework, the effect is not statistically detectable in 2nd grade but becomes detectable by 5th grade. This developmental pattern is consistent with earlier work showing that gender stereotypes concerning computing emerge during childhood and increasingly influence interest and sense of belonging [22]. Our findings extend

this line of work by suggesting that measurable PEOPLE–THINGS preferences become detectable by approximately 5th grade.

Progression throughout education. Within compulsory schooling, the observed gender effect strengthened from 5th to 8th grade. Viewed together with previous studies from upper secondary and university computing education [5, 19], our findings are consistent with the possibility that gender-differential preferences become more pronounced throughout education. Although the university study measured authentic exercise choice rather than ratings of hypothetical activities, and is therefore not directly comparable to our study, the overall trend is remarkably consistent. This progression also aligns with findings that PEOPLE–THINGS orientations predict educational choices, choice of major, and retention in thing-oriented disciplines [37].

Our findings should not be interpreted as showing that gender-differential preferences first arise around 5th grade. Rather, they indicate that such preferences become empirically detectable at this stage within our experimental framework. Earlier differences may simply be more difficult to measure reliably among younger pupils [21]. Whether this reflects developmental changes, increasing influence of gender stereotypes, or improved measurability as pupils mature remains an open question.

Educational implications. Our findings suggest that thematic framing constitutes a meaningful design dimension for computing educators. Simply changing the thematic context of otherwise isomorphic educational activities was associated with measurable

differences in pupils' preferences while leaving the underlying technical content unchanged. Such adaptations therefore have the potential to broaden the appeal of computing activities without altering their intended learning outcomes. If such preferences influence pupils before educational pathways diverge, interventions during compulsory schooling have the potential to reach a substantially broader population than initiatives introduced later.

This interpretation is consistent with observations by Margolis and Fisher [20], who argue that many women are attracted to computing through opportunities to solve meaningful problems involving people. Likewise, our findings complement earlier work showing that seemingly superficial characteristics of learning environments can influence students' engagement with computing [4]. Together, these studies suggest that thematic framing constitutes one design dimension among several that educators can deliberately consider when designing inclusive computing activities.

Appeal of thematic framing. Beyond gender differences, we found considerable variation in the appeal of individual PEOPLE-oriented themes, suggesting that thematic framing constitutes a richer design space than a simple PEOPLE/THINGS dichotomy. Understanding which characteristics contribute most strongly to this appeal remains an important direction for future work. To facilitate such analyses, we include all twelve translated exercises in Figure 7 (see supplementary material for full-page figures) and the corresponding pupil rating distributions in Figure 5.

6 THREATS TO VALIDITY

This study was conducted in authentic primary and lower secondary school settings involving young pupils. Consequently, several threats to validity warrant explicit consideration.

Establishing gender. Pupils self-reported gender using a free-text field. Virtually all participants (815/822; 99%) reported either 'girl' or 'boy'. Two 'non binary' responses were omitted as described earlier. The resulting gender distribution (50.7% girls, 49.3% boys) closely matches the national average (49% girls, 51% boys).

Appropriate treatments. The PEOPLE and THINGS versions were technically identical and *syntactically isomorphic*, isolating thematic framing as the intended treatment. We further ensured age-appropriate wording; Table 3 shows a gradual increase in vocabulary complexity from 2nd to 8th grade.

Comprehension and response validity in younger pupils. Although younger children may find hypothetical statements more challenging [21], we mitigated this through simplified tasks, demonstration exercises, and pre-recorded audio support. Responses from younger pupils should therefore be interpreted with appropriate caution, but remain informative.

Task understanding. To reduce misunderstandings, the questionnaire included three demo exercises illustrating strong preference, weak preference, and indifference. Pre-recorded audio further reduced reading-related difficulties for 2nd and 5th grade pupils.

Self-stereotyping. Respondents may self-stereotype when their identity is made salient [32]. To mitigate this threat, gender was requested only at the end of the questionnaire. Consequently, incomplete responses without a gender entry could not be analysed.

Presentation order bias. To mitigate possible left-to-right response bias, the left/right ordering of the PEOPLE and THINGS

versions was alternated using two mirrored questionnaire variants distributed randomly across classes.

Classroom context. Pupils completed the questionnaire in their regular classroom setting, but were explicitly instructed to answer independently, not look at peers' responses, and informed that there were no right or wrong answers. No incentives or rewards were provided. Nevertheless, peer presence or other immediate classroom factors may have influenced responses.

Thematic confounds. Although each PEOPLE-THINGS pair was syntactically isomorphic, themes may differ in familiarity, plausibility, vocabulary, or perceived meaningfulness. Thus, preferences cannot exclusively be attributed to the PEOPLE-THINGS dimension.

Experimenter bias. For 2nd and 5th grade pupils, pre-recorded audio with neutral intonation avoided experimenter influence. One pupil related to an author was excluded to reduce familiarity bias.

Beyond hypothetical exercises. Because the participating pupils had limited computing experience, the study relied on hypothetical exercises. Nevertheless, previous work using authentic programming assignments reports comparable effects [19], suggesting that the findings may extend beyond hypothetical settings.

7 CONCLUSION

Our study provides empirical evidence that gender-differential preferences for computing themes (PEOPLE vs THINGS) are detectable in primary and lower secondary education. The effect is not statistically detectable in 2nd grade, is detectable by 5th grade (possibly earlier), and is again detectable (and stronger) in 8th grade.

We also find that the PEOPLE-version has a stronger overall appeal to girls in absolute terms than the THINGS-version. Boys express less pronounced preferences, but nevertheless exhibit an inclination towards PEOPLE for several of the exercises.

These findings extend earlier results from upper secondary and university computing education [5, 19] to younger pupils and suggest that thematic framing may matter considerably earlier in the educational pipeline:

IMPLICATION (PEOPLE versus THINGS framing):

PEOPLE-oriented task framing should also be considered in primary and lower secondary computing education, with gender-differential preferences already detectable by 5th grade.

This may be especially relevant for *elective* computing courses, where broader appeal may influence participation and, over time, the diversity of the educational pipeline.

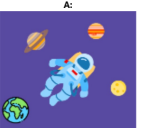
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2nd grade

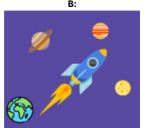
Imagine that you have Science & Technology where you need to read a text. You can choose between two texts:

A:



The text is about an **astronaut** who is sent into space and how the **astronaut** has to make sure not to get **stuck** on **his** journey.

B:

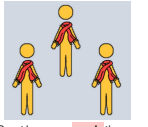


The text is about a **spaceship** that is sent into space and how the **spaceship** has to make sure not to get **stuck** on **its** journey.

2A

Imagine that you and your class need to practice counting during a Mathematics class. You can choose between two exercises:

A:



Count how many **people** there are in the store.

B:

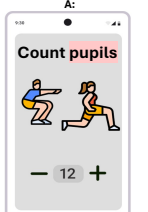


Count how many **packages** there are in the store.

2B

Imagine that your school has to make a new App for Sports class. Your class needs to help draw what the new App should look like. You can choose between two Apps:

A:



An App for keeping track of how many **pupils** are **present** at the Sports class. You are free to decide what it should look like. It could for instance look like this.

B:



An App for keeping track of how many **balls** are **used** at the Sports class. You are free to decide what it should look like. It could for instance look like this.

2C

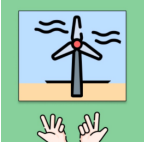
Imagine that you and your class need to practice counting during a Mathematics class. You can choose between two exercises:

A:



Count how many times the **heart** of the **child** **beats**.

B:



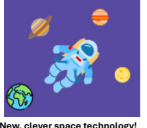
Count how many times the **wing** of the **windmill** **turns**.

2D

5th grade

Imagine that you have to make an exercise in school where you have to write a summary of a news article that you have read. You can choose between two news articles:

A:



New, clever space technology! Researchers have made a new system that can keep an eye on the **health** of **astronauts**. The system can suggest the best **exercises** for the **astronaut** so that the **astronaut** can stay **healthy** in space. Read more.

B:

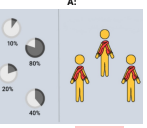


New, clever space technology! Researchers have made a new system that can keep an eye on the **battery** of **spaceships**. The system can suggest the best **configurations** for the **spaceship** so that the **spaceship** can stay **charged** in space. Read more.

5A

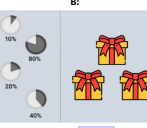
Imagine that your school needs to make a new App for use during Sports class. Your class needs to help design what the new App should look like. You can choose between two Apps:

A:



Count how many **customers** are in the store and calculate the percentage of **customers** in the **queue**.

B:

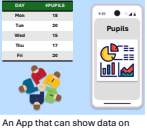


Count how many **packages** are in the store and calculate the percentage of **packages** on the **counter**.

5B

Imagine that you need to make an exercise during Mathematics class where you need to collect data from a store and calculate percentages from the data. You can choose between two exercises:

A:



An App that can show data on how many **pupils** are **present** at Sports class. The App should visualize data you have received from your teacher. The App could for example show a piechart or a table of the distribution of **pupils** at Sports class on different days.

B:



An App that can show data on how many **tools** are **used** at Sports class. The App should visualize data you have received from your teacher. The App could for example show a piechart or a table of the distribution of **tools** at Sports class on different days.

5C

8th grade

Imagine that your Physics teacher has taught your class some new systems that are used in space. You need to make an exercise which is about writing an article about one of the systems. You can choose between two exercises:

A:



There is a new system which can measure the **health** of the **astronaut**. Sensors collect data on **strength** and **endurance** and the system analyses this data to figure out the **best exercises** for the astronaut's body to **maintain health**. Write an article about this system.

B:



There is a new system which can measure the **powerlevel** of the **spaceship**. Sensors collect data on **environment** and **weather** and the system analyses this data to figure out the **best positions** for the **spaceship's** solarcells to **recharge batteries**. Write an article about this system.

8A

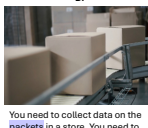
Imagine that your class in a Mathematics class need to make different exercises based on data that you collect from a store. You can choose between two exercises:

A:



You need to collect data on the **customers** in a store. You need to make different diagrams and tables to show your observations. It could, for example, be how many **customers** are in the store, how many **products** customers buy, on average, or what time of day, there are most **customers** in the store.

B:

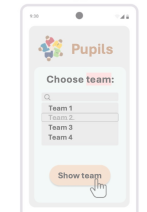


You need to collect data on the **packages** in a store. You need to make different diagrams and tables to show your observations. It could, for example, be how many **packages** are in the store, how many **grams** packages weigh, on average, or what time of day, there are most **packages** on the counter.

8B

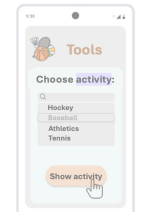
Imagine that your class has had a school subject where you have learned how to program Apps. You need to make a project which is about programming an App. You can choose between two projects:

A:



An App where you can pick a **team** and get an overview of **pupils** on the **team**. You can choose how you would like to design the App. It could, for example, look like this.

B:



An App where you can pick an **activity** and get an overview of **tools** for the **activity**. You can choose how you would like to design the App. It could, for example, look like this.

8C[†]

Imagine that your class has had a school subject where you have learned how to program Apps. You need to make a project which is about programming an App. You can choose between two projects:

A:



You should program an App that provides information on the **pulse** of a **pupil** as logged by a sensor. There should be a graph which shows the **pulse** of the **pupil** over time (measured in **beats per minute**). It should also provide information about the average **pulse** when at rest and average **pulse** when in motion.

B:



You should program an App that provides information on the **production** of a **windmill** as logged by a sensor. There should be a graph which shows the **production** of the **windmill** over time (measured in **kilowatt per hour**). It should also provide information about the average **production** when at rest and average **production** when in motion.

8D

Figure 7: All twelve exercises used in the study. This overview is intended primarily for on-screen zoomed-in viewing; we refer the reader to the online supplementary material for full-page figures. Differences between the PEOPLE and THINGS versions have been highlighted in red and blue, respectively. [†]Results from Exercise 8c were excluded (see Section 4).

REFERENCES

- [1] Alan Agresti. 2012. *Categorical Data Analysis*. Wiley.
- [2] Geneviève Allaire-Duquette, Pierre Chastenay, Thérèse Bouffard, Simon A. Bélanger, Olivier Hernandez, Mohamed Amine Mahhou, Patrick Giroux, Sophie McMullin, and Estelle Desjarlais. 2022. Gender Differences in Self-efficacy for Programming Narrowed After a 2-h Science Museum Workshop. *Canadian Journal of Science, Mathematics and Technology Education* 22, 1 (2022), 87–100. <https://doi.org/10.1007/s42330-022-00193-7>
- [3] World Bank. 2023. Urban population (% of total population) - Denmark. <https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS?locations=DK>
- [4] Sapna Cheryan, Victoria C. Plaut, Paul G. Davies, and Claude M. Steele. 2009. Ambient Belonging: How Stereotypical Cues Impact Gender Participation in Computer Science. *Journal of Personality and Social Psychology* 97, 6 (2009), 1045–1060. <https://doi.org/10.1037/a0016239>
- [5] Ingrid Maria Christensen, Melissa Høegh Marcher, Pawel Grabarczyk, Therese Graversen, and Claus Brabrand. 2021. Computing Educational Activities Involving People Rather Than Things Appeal More to Women (Recruitment Perspective). In *ICER 2021: ACM Conference on International Computing Education Research, Virtual Event, USA, August 16–19, 2021*. ACM, 127–144. <https://doi.org/10.1145/3446871.3469758>
- [6] Rune H. B. Christensen. 2023. *ordinal—Regression Models for Ordinal Data*. <https://CRAN.R-project.org/package=ordinal> R package version 2023.12-4.1.
- [7] Council of European Professional Informatics Societies (CEPIS). 2026. *SDD Report: Digital Skills Progress Continues, ICT Talent Shortage Still Present*. CEPIS. <https://cepis.org/sdd-report-digital-skills-progress-continues-ict-talent-shortage-still-present/>
- [8] Statistics Denmark. 2025. Primary School. <https://www.dst.dk/dk/Statistik/emner/uddannelse-og-forskning/fuldtidsuddannelser/grundskole>
- [9] Eurostat. 2024. *ICT specialists in employment*. Eurostat. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=ICT_specialists_in_employment
- [10] Eurostat. 2025. *ICT education - a statistical overview*. Eurostat. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=ICT_education_-_a_statistical_overview
- [11] Michelle Friend. 2017. Girls' Interest in Computing: Types and Persistence. In *Proceedings of 12th International Conference on Computer Supported Collaborative Learning (CSCCL)*, B. K. Smith, M. Borge, E. Mercier, and K. Y. Lim (Eds.). International Society of the Learning Sciences (ISLS), 71–78.
- [12] Pawel Grabarczyk, Alma Freiesleben, Amanda Bastrup, and Claus Brabrand. 2022. Computing Educational Programmes with more Women are more about People & less about Things. In *Proceedings of the 27th ACM Conference on Innovation and Technology in Computer Science Education Vol. 1 (Dublin, Ireland) (ITiCSE '22)*. Association for Computing Machinery, New York, NY, USA, 172–178. <https://doi.org/10.1145/3502718.3524784>
- [13] William Graziano, Meara Habashi, Demetra Evangelou, and Ida Ngambeki. 2012. Orientations and motivations: Are you a “people person,” a “thing person,” or both? *Motivation and Emotion* 36 (12 2012). <https://doi.org/10.1007/s11031-011-9273-2>
- [14] Shuchi Grover and Roy D. Pea. 2013. Computational Thinking in K–12: A Review of the State of the Field. *Educational Researcher* 42, 1 (2013), 38–43. <https://doi.org/10.3102/0013189X12463051>
- [15] Yasmin B. Kafai and Quinn Burke. 2014. *Connected Code: Why Children Need to Learn Programming*. The MIT Press, Cambridge, MA. <https://doi.org/10.7551/mitpress/9992.001.0001>
- [16] Richard A. Lippa. 1998. Gender-related individual differences and the structure of vocational interests: The importance of the people-things dimension. *Journal of Personality and Social Psychology* 74, 4 (1998), 996–1009. <https://doi.org/10.1037/0022-3514.74.4.996>
- [17] Richard A. Lippa. 1998. Gender Differences in Personality and Interests: When, Where, and Why? *Social and Personality Psychology Compass* 4, 11 (2010), 1098–1110. <https://doi.org/10.1111/j.1751-9004.2010.00320.x>
- [18] Linda Mannila, Anna Åkerfeldt, Susanne Kjällander, and Fredrik Heintz. 2022. Exploring Gender Differences in Primary School Programming. In *2022 IEEE Frontiers in Education Conference (FIE)*. Institute of Electrical and Electronics Engineers (IEEE), 1–9. <https://doi.org/10.1109/FIE56618.2022.9962482>
- [19] Melissa Høegh Marcher, Ingrid Maria Christensen, Pawel Grabarczyk, Therese Graversen, and Claus Brabrand. 2021. Computing Educational Activities Involving People Rather Than Things Appeal More to Women (CS1 Appeal Perspective). In *ICER 2021: ACM Conference on International Computing Education Research, Virtual Event, USA, August 16–19, 2021*. ACM, 145–156. <https://doi.org/10.1145/3446871.3469761>
- [20] Jane Margolis and Allan Fisher. 2002. *Unlocking the clubhouse: Women in computing*. MIT press.
- [21] Andrea Mari. 2024. Making conventional data collection more Child-friendly: Questionnaires with young students. *African Evaluation Journal* 12, 1 (2024), 690.
- [22] Allison Master, Andrew N. Meltzoff, and Sapna Cheryan. 2021. Gender stereotypes about interests start early and cause gender disparities in computer science and engineering. *Proceedings of the National Academy of Sciences of the United States of America* 118, 48 (2021), e2100030118. <https://doi.org/10.1073/pnas.2100030118>
- [23] Allison Master, Daijazi Tang, Desiree Forsythe, Taylor M. Alexander, Sapna Cheryan, and Andrew N. Meltzoff. 2023. Gender equity and motivational readiness for computational thinking in early childhood. *Early Childhood Research Quarterly* 64 (2023), 242–254. <https://doi.org/10.1016/j.ecresq.2023.03.004>
- [24] Catia Milesi, Laura Pérez-Felkner, Keith Brown, and Barbara Schneider. 2017. Engagement, persistence, and gender in computer science: Results of a smartphone ESM study. *Frontiers in Psychology* 8 (2017), 602. <https://doi.org/10.3389/fpsyg.2017.00602>
- [25] Ministry of Children and Education. 2021. Progress in International Reading Literacy Study (PIRLS 2021). <https://www.uvm.dk/internationalt-arbejde/internationale-undersogelser/pirls/pirls-2021>
- [26] Ministry of Children and Education. 2022. Programme for International Student Assessment (PISA2022). <https://www.uvm.dk/internationalt-arbejde/internationale-undersogelser/pisa/pisa-2022>
- [27] Ministry of Children and Education. 2024. Free Schools in Numbers. <https://www.uvm.dk/frie-grundskoler/om-skoler-og-formaal/frie-grundskoler-i-tal>
- [28] Ministry of Children and Education. 2025. Common Objectives for School Subjects. <https://www.uvm.dk/folkeskolen/fag-timetall-og-overgange/fag-emner-og-tvaergaende-temaer/faelles-maal/faelles-maal-for-folkeskolens-fag>
- [29] Maria Ong, Carol Wright, Lorelle Espinosa, and Gary Orfield. 2011. Inside the Double Bind: A Synthesis of Empirical Research on Undergraduate and Graduate Women of Color in Science, Technology, Engineering, and Mathematics. *Harvard Educational Review* 81, 2 (2011), 172–209.
- [30] Lori Postner, Gregory W Hislop, and Heidi J.C. Ellis. 2023. Humanitarian Applications Increase Interest and Motivation of Women in Computing. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (Toronto ON, Canada) (SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 416–422. <https://doi.org/10.1145/3545945.3569832>
- [31] Andreas Brøns Riise. 2023. Educator and Expert: Here are the Challenges in the Government Plan for Technology Comprehension. <https://www.folkeskolen.dk/laerer-og-ekspert-her-er-udfordringerne-i-regeringens-plan-for-teknologiforstaelse/>
- [32] Stacey Sinclair, Curtis D Hardin, and Brian S Lowery. 2006. Self-stereotyping in the context of multiple social identities. *Journal of personality and social psychology* 90, 4 (2006), 529.
- [33] Rong Su and James Rounds. 2015. All STEM fields are not created equal: People and things interests explain gender disparities across STEM fields. *Frontiers in psychology* 6 (2015), 189. <https://doi.org/10.3389/fpsyg.2015.00189>
- [34] Rong Su, James Rounds, and Patrick Ian Armstrong. 2009. Men and things, women and people: A meta-analysis of sex differences in interests. *Psychological Bulletin* 135, 6 (2009), 859–884. <https://doi.org/10.1037/a0017364>
- [35] Grete Katrine Teilmann. 2025. Patient's Handbook. <https://www.sundhed.dk/borger/patienthaandbogen/boern/sygdomme/vaekst-og-udvikling/pubertet/Puberty>
- [36] Jeannette M. Wing. 2006. Computational Thinking. *Commun. ACM* 49, 3 (2006), 33–35. <https://doi.org/10.1145/1118178.1118215>
- [37] Ngam Woodcock, William G Graziano, Sara E Branch, Meara M Habashi, Ida Ngambeki, and Demetra Evangelou. 2013. Person and thing orientations: Psychological correlates and predictive utility. *Social psychological and personality science* 4, 1 (2013), 116–123.